RECOVERING THE TAIL OF A GRADED MODULE OVER A GENERAL NOETHERIAN GRADED RING: FOR A NOETHERIAN GRADED, D THE LCM OF DEGREES OF GENERATORS OVER A_0, M A FINITE GRADED MODULE, X = PROJ(A), K IN Z: N' = BIGOPLUS_{N >= K} H^0(X, M(N)^TILDE) AND N = BIGOPLUS_{N >= K} H^0(X, M^TILDE(N)) ARE FINITE A-MODULES, THERE ARE CANONICAL MAPS N -> N' AND (K SMALL) M -> N', M_N -> N'_N IS AN ISOMORPHISM FOR N » 0, AND N_N -> N'_N IS AN ISOMORPHISM WHENEVER D DIVIDES N.

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4297-4314 of sources/stacks-coherent/pinned-original-source.tex; proof span is lines 4316-4414, the complete author proof (97 body lines).

Editorial note. Navigation stubs: 9 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Card-surface disclosure per the tightened ruling: the identity of the composite N^tilde -> M^tilde -> N'^tilde with the constructed isomorphism is the author's named omission ('proof omitted; hint'), and shift-compatibility of N' is called immediate from the construction; all six enumerated claims carry their proofs in the span.

Editorial note. Span computed from the \label marker through the enclosing \end{lemma}; precard_gate RESULT=OK (probe L9999).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; every pointer named here occurs verbatim in the retained statement/proof text.

Source reference. Original lemma 07Z4: sources/stacks-coherent/pinned-original-source.tex, source lines 4414–4414. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07Z2: sources/stacks-coherent/pinned-original-source.tex, source lines 4414–4414. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:coherent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00JW: sources/stacks-coherent/pinned-original-source.tex, source lines 4414–4414. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01MR: sources/stacks-coherent/pinned-original-source.tex, source lines 4414–4414. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01MQ: sources/stacks-coherent/pinned-original-source.tex, source lines 4414–4414. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01M7: sources/stacks-coherent/pinned-original-source.tex, source lines 4414–4414. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01MU: sources/stacks-coherent/pinned-original-source.tex, source lines 4414–4414. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B5Q: sources/stacks-coherent/pinned-original-source.tex, source lines 4237–4255. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AG5: sources/stacks-coherent/pinned-original-source.tex, source lines 4414–4414. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let A be a Noetherian graded ring and let d be the lcm of generators of A over A_0. Let M be a finite graded A-module. Set $X = Proj(A)$ and let $\widetilde{M}$ be the quasi-coherent $\mathcal{O}_X$-module on X associated to M. Let $k \in \mathbf{Z}$.*

(1) *$N' = \bigoplus_{n \geq k} H^0(X, \widetilde{M(n)})$ is a finite A-module,*
(2) *$N = \bigoplus_{n \geq k} H^0(X, \widetilde{M}(n))$ is a finite A-module,*
(3) *there is a canonical map $N \to N'$,*
(4) *if k is small enough there is a canonical map $M \to N'$,*
(5) *the map $M_n \to N'_n$ is an isomorphism for $n \gg 0$,*
(6) *$N_n \to N'_n$ is an isomorphism for $d|n$.*

Proof. The map $N \to N'$ in (3) comes from Constructions, Equation (01MQ) by taking global sections.

By Constructions, Equation (01MR) there is a map of graded A-modules $M \to \bigoplus_{n \in \mathbf{Z}} H^0(X, \widetilde{M(n)})$. If the generators of M sit in degrees $\geq k$, then the image is contained in the submodule $N' \subset \bigoplus_{n \in \mathbf{Z}} H^0(X, \widetilde{M(n)})$ and we get the map in (4).

By Algebra, Lemmas 00JW and 07Z4 the ring A_0 is Noetherian and A is generated over A_0 by finitely many elements $f_1, \ldots, f_r$ homogeneous of positive degree. Let $d = \text{lcm}(\deg(f_i))$. Then we see that (6) holds for example by Constructions, Lemma 01MU.

Because M is a finite A-module we see that $\widetilde{M}$ is a finite type $\mathcal{O}_X$-module, i.e., a coherent $\mathcal{O}_X$-module. Thus part (2) follows from Lemma 0B5Q.

We will deduce (1) from (2) using a trick. For $q \in \{0, \ldots, d-1\}$ write

$$^qN = \bigoplus\nolimits_{n+q \geq k} H^0(X, \widetilde{M(q)}(n))$$

By part (2) these are finite A-modules. The Noetherian ring A is finite over $A^{(d)} = \bigoplus_{n \geq 0} A_{dn}$, because it is generated by f_i over $A^{(d)}$ and $f_i^d \in A^{(d)}$. Hence ^qN is a finite $A^{(d)}$-module. Moreover, $A^{(d)}$ is Noetherian (follows from Algebra, Lemma 07Z2). It follows that the $A^{(d)}$-submodule $^qN^{(d)} = \bigoplus_{n \in \mathbf{Z}} {}^qN_{dn}$ is a finite module over $A^{(d)}$. Using the isomorphisms $\widetilde{M(dn+q)} = \widetilde{M(q)}(dn)$ we can write

$$N' = \bigoplus\nolimits_{q \in \{0,\ldots,d-1\}} \bigoplus\nolimits_{dn+q \geq k} H^0(X, \widetilde{M(q)}(dn)) = \bigoplus\nolimits_{q \in \{0,\ldots,d-1\}} {}^qN^{(d)}$$

Thus N' is finite over $A^{(d)}$ and a fortiori finite over A. Thus (1) is true.

Let K be a finite A-module such that $\widetilde{K} = 0$. We claim that $K_n = 0$ for $d|n$ and $n \gg 0$. Arguing as above we see that $K^{(d)}$ is a finite $A^{(d)}$-module. Let $x_1, \ldots, x_m \in K$ be homogeneous generators of $K^{(d)}$ over $A^{(d)}$, say sitting in degrees $d_1, \ldots, d_m$ with $d|d_j$. For each i and j there exists an $n_{ij} \geq 0$ such that $f_i^{n_{ij}} x_j = 0$ in $K_{d_j + n_{ij}}$: if not then $x_j/f_i^{d_i/\deg(f_i)} \in K_{(f_i)}$ would not be zero, i.e., $\widetilde{K}$ would not be zero. Here we use that $\deg(f_i)|d|d_j$ for all i, j. We conclude that K_n is zero for n with $d|n$ and $n > \max_j(d_j + \sum_i n_{ij} \deg(f_i))$ as every element of K_n is a sum of terms where each term is a monomials in the f_i times one of the x_j of total degree n.

To finish the proof, we have to show that $M \to N'$ is an isomorphism in all sufficiently large degrees. The map $N \to N'$ induces an isomorphism $\widetilde{N} \to \widetilde{N'}$ because on the affine opens $D_+(f_i) = D_+(f_i^d)$ the corresponding modules are isomorphic: $N_{(f_i)} \cong N_{(f_i^d)} \cong N'_{(f_i^d)} \cong N'_{(f_i)}$ by property (6). By Properties, Lemma 0AG5 we have a canonical isomorphism $\widetilde{N} \to \widetilde{M}$. The composition $\widetilde{N} \to \widetilde{M} \to \widetilde{N'}$ is the isomorphism above (proof omitted; hint: look on standard affine opens to check this). Thus the map $M \to N'$ induces an isomorphism $\widetilde{M} \to \widetilde{N'}$. Let $K = \mathrm{Ker}(M \to N')$ and $Q = \mathrm{Coker}(M \to N')$. Recall that the functor $M \mapsto \widetilde{M}$ is exact, see Constructions, Lemma 01M7. Hence we see that $\widetilde{K} = 0$ and $\widetilde{Q} = 0$. By the result of the previous paragraph we see that $K_n = 0$ and $Q_n = 0$ for $d|n$ and $n \gg 0$. At this point we finally see the advantage of using N' over N: the functor $M \rightsquigarrow N'$ is compatible with shifts (immediate from the construction). Thus, repeating the whole argument with M replaced by $M(q)$ we find that $K_n = 0$ and $Q_n = 0$ for $n \equiv q \bmod d$ and $n \gg 0$. Since there are only finitely many congruence classes modulo n the proof is finished. $\square$